MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

SECOND SEMESTER PRACTICAL EXAMINATION

(2024 ADMISSION ONWARDS)

MG2MDCCSC100- Data Visualization using PYTHON

Duration of the Examination: 1.5 Hrs. Maximum Marks: 35

List of Questions

1.Program to create a simple line plot using matplotlib.

2. Create a line plot showing the GDP growth of a country over 10 years.Customize the plot with appropriate labels for the axes,a title, and a legend.
3. Create a bar plot using matplotlib.
4. Create a horizontal bar plot using matplotlib.
5. Program to create a bar plot using labels, legends in matplotlib.
6. Create a bar plot displaying the average monthly income of workers in different industries(eg. agriculture,IT,Healthcare).
7. Program to create a scatter plot using matplotlib.
8. Plot the relationship between population growth and unemployment rate for five countries. Use different colours for the points.
9. Program to create a sub plot using matplotlib.

10.Program to create a figure with multiple subplots using matplotlib.

11.Program to create a line plot using seaborn.

12.Program to create a scatter plot using seaborn.

13.Program to create a bar plot using seaborn.

14.Program to create a histogram using seaborn.

15.Program to create a box plot using seaborn.

16.Program to create a box plot with a theme using seaborn.

17.Use different seaborn themes(eg, "darkgrid","whitegrid") to plot a histogram of household incomes.

18.Program to create a multi-grid facetGrid using seaborn.

19.Program to create a multi-grid PairGrid using seaborn.

20. Program to create a multi-grid JointGrid using seaborn.
21. Program to create a stripplot in categorical plot using seaborn.
22. Program to create a Boxplot in categorical plot using seaborn.
23. Program to create a Violinplot in categorical plot using seaborn.
24. Create a bar plot showing the average expenditure on food, education, and healthcare by income groups(low, middle ,high) in categorical plot.
25. Program to create a Swarmplot in categorical plot using seaborn.
26. Program to create a combined Swarm+Violin plot using seaborn.
27. Create an interactive line chart showing the import and export data of a country over a decade using Plotly
28. Program to create a Plotly in simple dashboard creation.